

PROJECT REPORT

Simulating the Oscillation of an Aircraft Wing Under a Dynamic Angle of Attack

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1. Abstract

This project simulates the oscillation of a simplified aircraft wing model under aerodynamic loading and structural damping. A Python-based numerical model computes the wing's position, velocity, and the aerodynamic lift and drag acting on it over a 20-second flight window, using Euler integration to solve the underlying equations of motion at each 0.01-second time step. Lift and drag are modeled as functions of a dynamically changing angle of attack (AoA), including a stall condition above 15° . A real-time, three-panel animation visualizes wing position, lift force, and drag force simultaneously, giving an intuitive view of how aerodynamic forces and structural damping interact as the oscillation decays.

2. Introduction & Objectives

Aircraft wings are not rigid: under aerodynamic loading, a wing behaves like a damped spring-mass system, oscillating and gradually settling as structural damping dissipates energy. That oscillation is directly coupled to the aerodynamic forces acting on the wing, which themselves depend on the angle of attack, which changes as the wing moves. Understanding this coupled behavior, structural dynamics feeding back into aerodynamic loading and vice versa, is foundational to topics like flutter and aeroelastic stability, even before high-fidelity CFD or wind-tunnel testing enters the picture.

This project builds a lightweight, first-principles model of that coupling, with the following objectives:

- Model wing motion as a damped mass-spring system, driven by aerodynamic lift and drag that vary with a dynamically computed angle of attack.
- Represent stall behavior realistically, by capping lift generation once AoA exceeds a defined threshold (15°), reflecting real aerodynamic stall onset.
- Solve the resulting equations of motion numerically using Euler integration, at a fine enough time resolution (0.01 s, 100 Hz) to resolve the oscillation clearly.
- Visualize the simulation in real time, so the relationship between wing position, lift, and drag over time is intuitive rather than just numerical output.
- Keep the model's physical parameters (mass, stiffness, damping) configurable, so the same framework can be used to explore multiple flight/structural configurations.

3. Related Work

Two recent studies informed the framing of this project. Zhu and Jiang (2023) examined how aerodynamic interference between a rigid and an elastic airfoil affects transonic flutter, showing that interference can trigger earlier shock formation and widen the flutter boundary, underscoring how sensitive flutter behavior is to aerodynamic coupling. Separately, a 2023 study on tandem-wing aircraft used CFD to show a nonlinear relationship between wingspan and longitudinal dynamic stability, driven by aerodynamic coupling between the front and rear wings. Both studies reinforced the same underlying idea this project simplifies into a tractable model: aerodynamic forces and structural response are tightly coupled, and that coupling is where the interesting (and consequential) behavior lives.

4. Methodology

The wing was modeled as a single-degree-of-freedom mass-spring-damper system, with aerodynamic lift and drag layered on top as forcing terms that depend on the system's own motion, creating a feedback loop between structural response and aerodynamic loading.

Physical Parameters

Parameter	Role
Mass, stiffness, damping	Define the structural (spring-mass-damper) behavior of the wing
Lift & drag coefficients (C_{l_0} , C_{d_0})	Baseline aerodynamic coefficients at zero angle of attack
Stall threshold (15°)	Angle of attack beyond which lift generation is capped, modeling stall
Initial position & velocity	Starting condition for the oscillation
Time step (0.01 s) & duration (20 s)	Numerical resolution and length of the simulated flight window

Aerodynamic Force Model

At each time step, the angle of attack was derived from the wing's current position and velocity. Below the stall threshold, the lift coefficient was modeled to vary with AoA following a $\sin(2\alpha)$ relationship, and the drag coefficient followed a $\cos^2(\alpha)$ relationship, capturing the characteristic rise in lift and fall in drag efficiency as AoA increases from zero. Once AoA exceeded 15°, lift was set to zero and drag held at its baseline value, reflecting the loss of lift associated with stall. Lift and drag forces were then computed from these coefficients and the current velocity at every step.

Numerical Integration

The coupled equations of motion, combining spring and damping forces with the computed aerodynamic lift, were advanced in time using Euler's method: acceleration was computed from the net force at each step, then used to update velocity, and velocity used to update position, over 2,000 time steps spanning the 20-second simulation window. Damping was scaled to increase gradually across the simulation, so the model could represent up to five configurable oscillation cycles decaying over time, rather than a fixed, unchanging damping rate.

Visualization

Rather than only inspecting the final numerical output, the simulation was paired with a real-time, three-panel animation, one panel each for wing position, lift force, and drag force, all plotted against time and updated frame-by-frame as the simulation advanced. This made it possible to see, directly, how a spike in lift or drag corresponded to a specific point in the wing's oscillation, rather than having to cross-reference separate static plots.

5. Results

The simulation produced a damped oscillation in wing position, with lift and drag forces tracking the changing angle of attack throughout the 20-second window. As expected from the $\sin(2\alpha)/\cos^2(\alpha)$ force model, lift rose and fell in a cyclical pattern tied to the oscillation itself, while drag remained comparatively steadier except where the AoA approached the stall threshold, at which point lift was suppressed and the system relied on damping and drag alone to bring the oscillation back down.

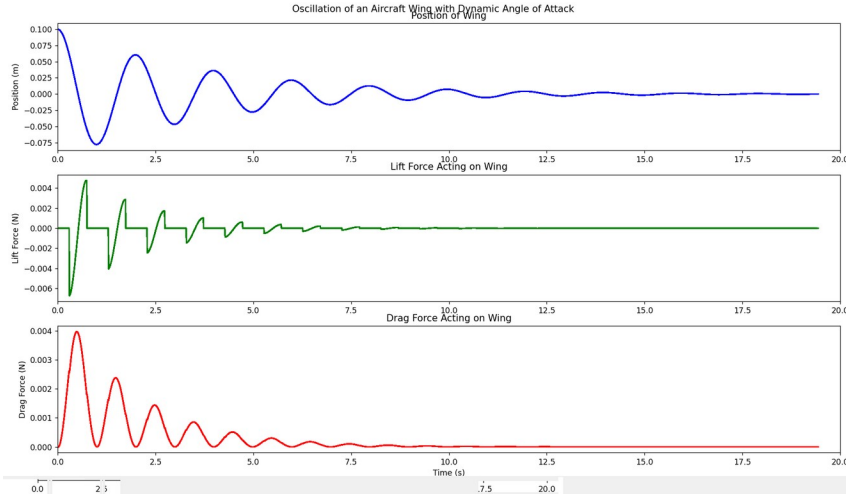


Fig. 1 — Simulation output: wing position (top), lift force (middle), and drag force (bottom) over the 20-second simulated flight window.

Because mass, stiffness, and damping were kept as configurable inputs, the same model could be re-run across more than five different structural/flight configurations without changing the underlying force model, making it a reusable tool for exploring how structural parameters affect oscillation behavior, not just a single fixed test case.

6. Conclusion

This project shows that even a simplified, first-principles model, a single-degree-of-freedom structural system coupled with an angle-of-attack-dependent aerodynamic force model, is enough to reproduce the qualitative behavior expected of a real wing under aerodynamic loading: damped oscillation, cyclical lift and drag, and a clear loss of lift once stall is reached. The real-time visualization made that coupling legible in a way a static result table would not have, and the configurable parameter set means the same framework extends naturally to comparing multiple structural configurations. As a foundation, this model is a useful stepping stone toward more advanced flutter and aeroelastic analysis, without requiring the computational cost of a full CFD or mid-fidelity aerodynamic solver at this stage.

7. References

Zhu, Y. & Jiang, P. Effects of angle of attack on transonic flutter of a supercritical wing. (2023). <https://arc.aiaa.org/doi/10.2514/1.J062970>

Analysis of longitudinal dynamic stability of tandem wing aircraft. (2023). <https://www.emerald.com/insight/content/doi/10.1108/AEAT-11-2022-0328/full/html>